Pick's formula (12.3.2.80)

A. Engel — Problem-Solving Strategies · 1998

Pick proved a simple formula for the area $f(P)$ of any lattice polygon P :

$$A = f(P) = i + \frac{b}{2} - \frac{1}{2}.$$

Here i and b are the numbers of interior and boundary points, respectively. We leave the the proof to you, but we give you the steps leading to a proof.

(a) Prove the formula for any lattice rectangle with sides p and q.

(b) Prove the formula for a right triangle with one horizontal and one vertical side.

(c) Pick's formula assigns a number $f(P)$ to any polygon P. Prove that the function f is additive, i.e., if P_1, P_2 are polygons with a common boundary, then $f(P_1 \cup P_2) = f(P_1) + f(P_2)$.

(d) Show that $f(P)$ gives the correct area for any lattice triangle P.

(e) Finally, show that $f(P)$ is the area of any simple lattice polygon P.